

Quantum Computing LEI 25/26

Introduction to quantum algorithms with PennyLane

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Quantum programming languages ?!

Not really, Quantum software development Kits (SDKs)

SDKs usually in Python/C++

Classical instructions that build circuits
Enabling simulation to hardware execution

Quantum technology companies



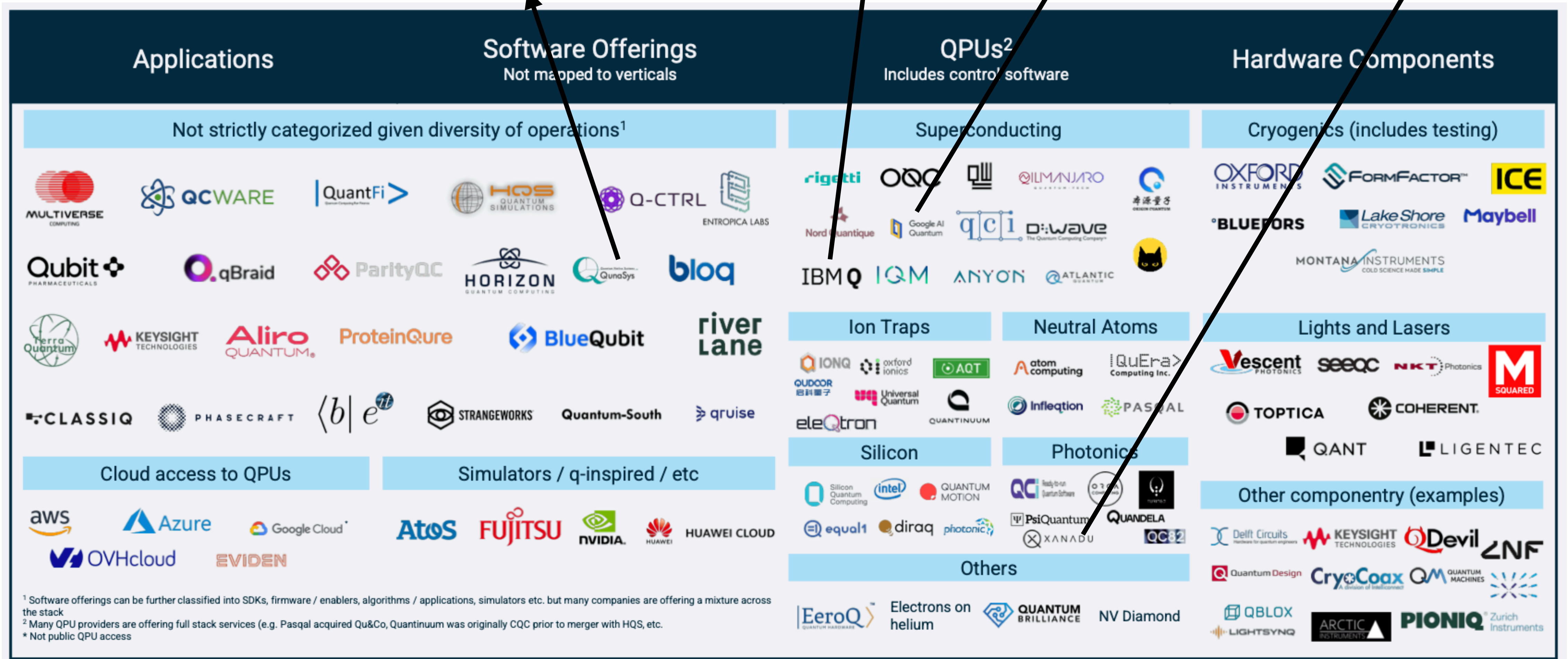
$|Q\rangle$



Qiskit

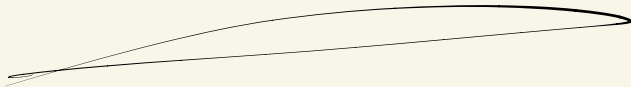


Cirq



What is a qubit ?!

classical bit



Unit of information
classical computer

bit 0

bit 1

classical bit

Unit of information
classical computer

bit 0

bit 1

How do we manipulate bits ?
logical gates

0 —▷ 1 NOT

1 ≡ 1 — 1 AND

1 ≡ 1 — 0 NAND

1 ≡ 1 — 0 XOR

universal gate set for
classical computation
(only NANDs actually)

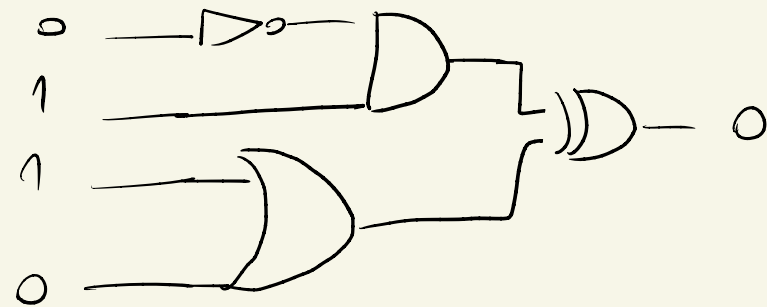
classical bit

Unit of information
classical computer

bit 0

bit 1

How do we manipulate bits ?
logical gates



Deterministic
computation

classical bit

Unit of information
classical computer

bit 0

bit 1

(deterministic)

Quantum bit

Unit of information
~~classical~~ computer
quantum

qubit $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

qubit $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

but

classical bit

Unit of information
classical computer

bit 0

bit 1

Quantum bit

Unit of information
~~classical~~ computer
quantum

$$\text{qubit } |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{qubit } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

but

$$\begin{aligned} \text{qubit } |\psi\rangle &= \alpha |0\rangle + \beta |1\rangle \\ &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \end{aligned} \quad \begin{array}{l} \text{linear} \\ \text{combination} \end{array}$$

The outcome of the quantum computation is
probabilistic

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

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Does it mean the qubit is a probabilistic bit?

The outcome of the quantum computation is
probabilistic

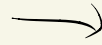
$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Does it mean the qubit is a probabilistic bit?

No, because $\alpha, \beta \in \mathbb{C}$ Amplitudes
can interfere

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$\alpha, \beta \in \mathbb{C}$$



probability of obtaining 0,1

$$P_{|0\rangle} = |\alpha|^2$$

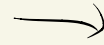
$$P_{|1\rangle} = |\beta|^2$$

$$P_{|0\rangle} + P_{|1\rangle} = 1$$

Always?

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha, \beta \in \mathbb{C}$$



probability of obtaining 0,1

$$P_{|0\rangle} = |\alpha|^2$$

$$P_{|1\rangle} = |\beta|^2$$

$$P_{|0\rangle} + P_{|1\rangle} = 1$$

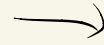
always?

The qubit is a complex vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ that by the end of the computation returns a classical bit w/ probabilities $|\alpha|^2, |\beta|^2$ s.t

$$|\alpha|^2 + |\beta|^2 = 1$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha, \beta \in \mathbb{C}$$



probability of obtaining 0,1

$$P_{|0\rangle} = |\alpha|^2$$

$$P_{|1\rangle} = |\beta|^2$$

$$P_{|0\rangle} + P_{|1\rangle} = 1$$

Always?

How do we manipulate the qubit?

Quantum logic gates that...

Preserve the length of the vector $|\alpha|^2 + |\beta|^2 = 1$

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Unitary matrices act as our logic gates

$$U^\dagger U = I$$

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Unitary matrices act as our logic gates

$$U^\dagger U = I \quad U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad U^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

quantum gates $\rightarrow$ quantum circuits

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$\rightarrow$

$$|0\rangle \text{ --- } \boxed{U} \text{ ---}$$

$$|1\rangle \text{ --- } \boxed{U} \text{ ---}$$

But crucially quantum gates can receive superpositions as input

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

~~~~~

$$|\psi\rangle \rightarrow \boxed{U} \rightarrow |\psi'\rangle$$

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$U$  is a linear operator  
(acts in each term of the superposition)

$$|\psi'\rangle = U|\psi\rangle =$$

$$= U \begin{pmatrix} \alpha \\ \beta \end{pmatrix} =$$

$$= U \begin{pmatrix} \alpha \\ 0 \end{pmatrix} + U \begin{pmatrix} 0 \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}$$

Ex : Verify if the following matrices  
are valid quantum gates

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = A^+ \quad AA^+ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = B^+ \quad BB^+ = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = C^+ \rightarrow CC^+ = \underline{I}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

—  $\boxed{X}$  —

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

—  $\boxed{Y}$  —

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

—  $\boxed{Z}$  —

Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\boxed{X}$                        $\boxed{Y}$                        $\boxed{Z}$

Pauli matrices

Ex: Verify what they do to our computational basis states  $|0\rangle, |1\rangle$

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

~~~~~ NOT gate

$$Y|0\rangle = i|1\rangle$$

$$Y|1\rangle = -i|0\rangle$$

~~~~~ complex NOT gate

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = |1\rangle$$

~~~~~ phase shift  $|1\rangle$


what about

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$H = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} = H^\dagger \rightarrow HH^\dagger = I$$

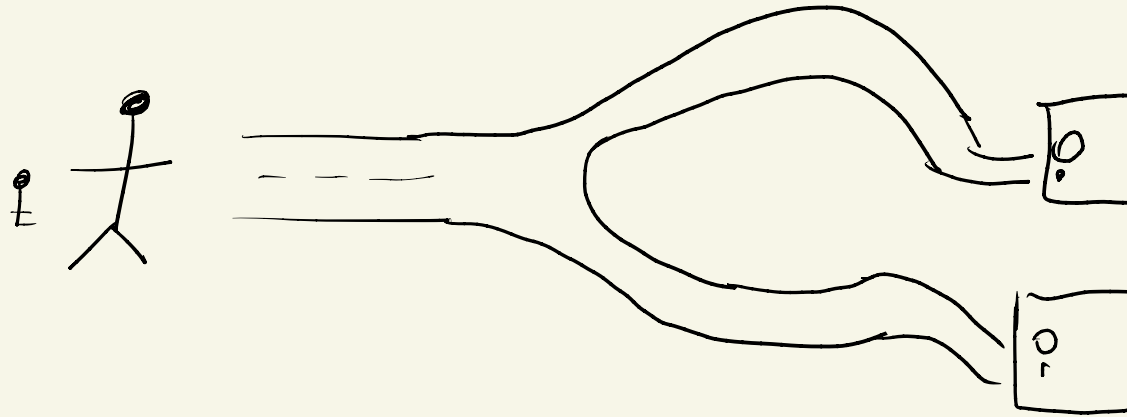
$$H|0\rangle = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$H|1\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Hadamard gate
creates superposition



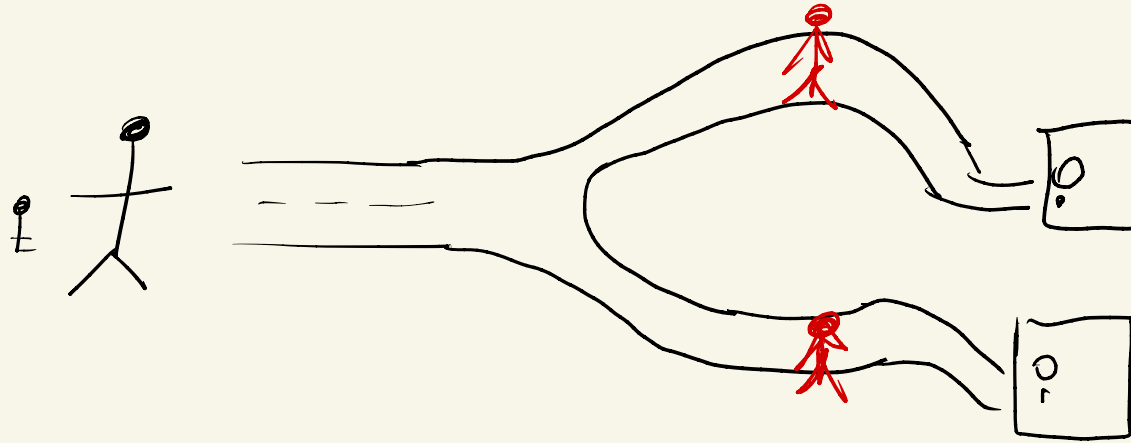
Imagine your favorite game character



Road 0, 1

classical bot would test
one road at a time

Imagine your favorite game character



Hadamards create superpositions so
the quantum bot experiences
both roads

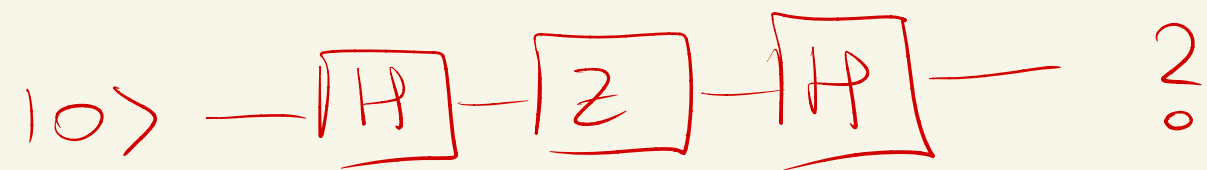
Awesome! Is it enough?

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Quantum computation would return
the correct road w/ probability
 $1/2$!

So, superposition alone is not enough !

Ex:



$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$= \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right)$$

$$= \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

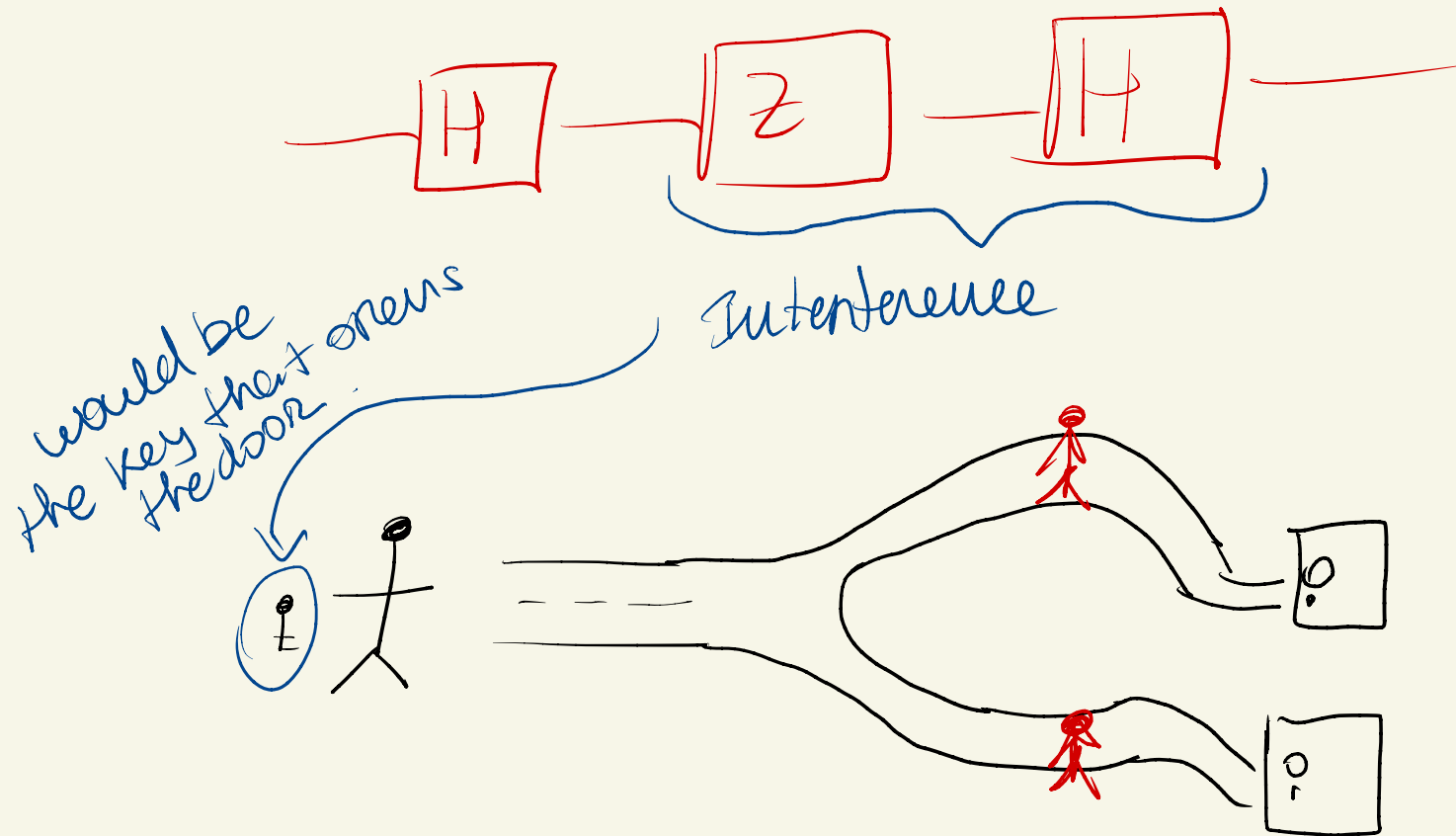
$$H \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right)$$

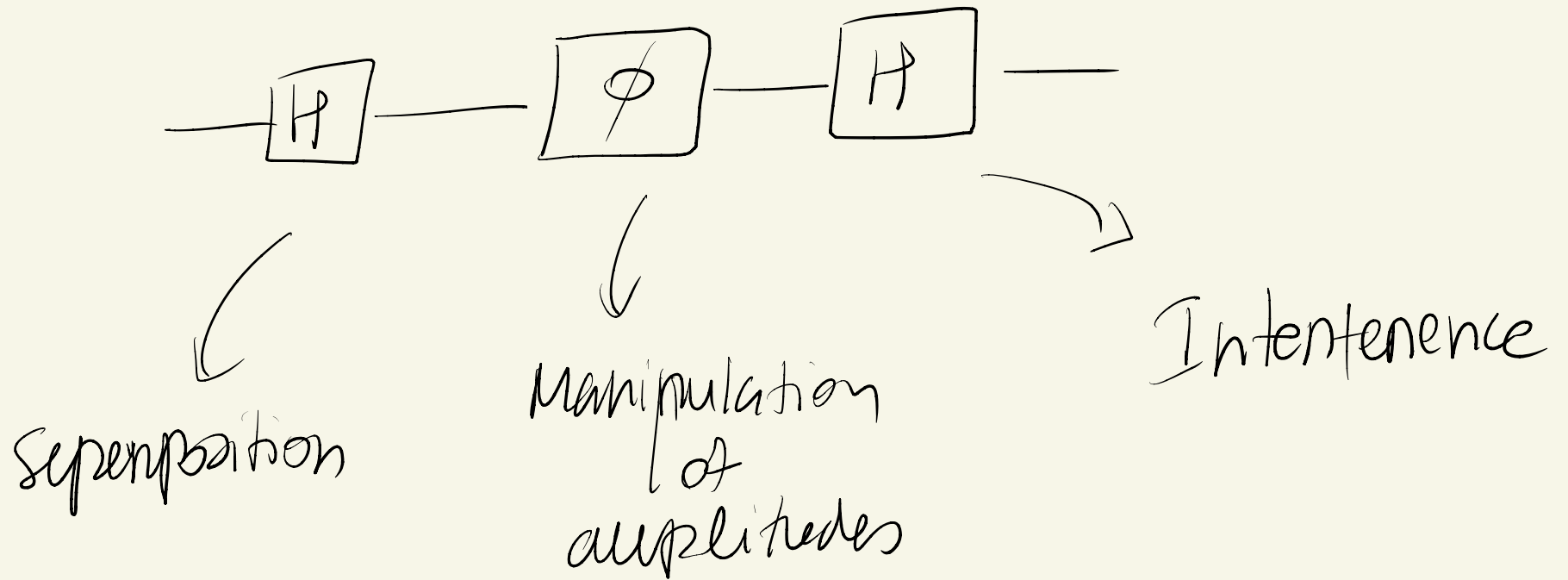
$$- \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right)$$

$$= |1\rangle$$

Amplitudes interfered s.t
amplitude $|0\rangle$ is zero!



A quantum algorithm will be
in general a computation



gate set so far

$\{H, X, Y, Z, Y$

Is it enough?

$10 > \text{---} \boxed{11}$

$10 > \text{---} \boxed{X}$

$10 > \text{---}$

$\begin{matrix} 0 \\ 1 \end{matrix} \text{---} \bigcup \text{---} 1$

There is certainly
many more single qubit
gates but

multiple qubit

computation

require

multi qubit gates